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Devices, Systems, And Methods For Catching Projectiles

Abstract

The disclosure relates to a system for visually detecting projectiles to thereby confirm that the projectiles are not trapped in a conduit. The system includes a housing defining an inlet. A trap is positioned within the housing. The trap is moveable about and between a catching configuration and a releasing configuration. An actuator is configured to move the trap about and between the catching configuration and the releasing configuration. The housing has a window that permits visual detection of the trap from a position outside of the housing.

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Background/Summary

FIELD
[0001] The disclosure relates to devices, systems, and methods for catching projectiles. The disclosed devices and systems can be used for catching projectiles that have passed through a conduit such as a hydraulic hose. The disclosed devices and systems can provide optical confirmation that a projectile has passed through the conduit, indicating that the projectile is not stuck within the hose. Methods of using the devices and systems for catching projectiles are also provided.
BACKGROUND
[0002] A standard test for a conduit, such as a hydraulic hose, to confirm that the hose does not have any partial or complete blockages that would inhibit flow therethrough includes firing a pellet therethrough. If the pellet passes through the conduit, the conduit passes the test; if the pellet does not exit the conduit, the conduit fails the test. Typically, an operator watches to confirm whether the pellet has passed through the conduit and then handles the test pellet. Accordingly, the test is labor intensive. Some automated systems have been introduced, but they are unreliable.
[0003] It is therefore desirable to provide way of reducing the amount of labor required to perform the test while maintaining reliability.
SUMMARY OF EMBODIMENTS
[0004] The disclosure relates to systems and methods for visually detecting projectiles to thereby confirm that the projectiles are not trapped in a conduit.
[0005] In one aspect, a system includes a housing defining an inlet. A trap is positioned within the housing. The trap is moveable about and between a catching configuration and a releasing configuration. An actuator is configured to move the trap about and between the catching configuration and the releasing configuration. The housing has a window that permits visual detection of the trap from a position outside of the housing.
[0006] In another aspect, a method includes coupling a first end of a conduit to a projectile launcher; positioning a second end of the conduit at the inlet of the housing; and propelling, using the projectile launcher, a pellet through the conduit.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0007] The invention will now be described, by way of example, with reference to the accompanying drawings in which:
[0008] FIG. **1** is a perspective view of an exemplary system for visually detecting projectiles as disclosed herein.
[0009] FIG. **2** is a side view of the exemplary system of FIG. **1**.
[0010] FIG. **3** is a perspective view of a housing and a trap of the exemplary system of FIG. **1**.
[0011] FIG. **4** is a perspective view of the housing and the trap of the exemplary system of FIG. **1**, with a portion of the housing removed to see internal components.
[0012] FIG. **5** is a side view of a housing and a trap as in FIG. **3**.
[0013] FIG. **6** is a partial side view of the housing and the trap as in FIG. **3**, with the trap in a catching configuration.
[0014] FIG. **7** is a partial side view of the housing and the trap as in FIG. **3**, with the trap in a releasing configuration.
[0015] FIG. **8**A is a schematic drawing an exemplary trap of a system as disclosed herein, with the trap in a releasing configuration. FIG. **8**B is a schematic drawing the exemplary trap of FIG. **8**A, with the trap in a releasing configuration.
[0016] FIG. **9** is a schematic diagram of an exemplary system for visually detecting projectiles as disclosed herein.

[0017] FIG. **10** is a schematic diagram of an exemplary system for visually detecting projectiles as disclosed herein.
[0018] FIG. **11** is a block diagram of an exemplary computing system comprising a computing device as disclosed herein.

DETAILED DESCRIPTION OF EMBODIMENTS

[0019] Before the present systems and methods are described, it is to be understood that the present disclosure is not limited to the particular processes, compositions, or methodologies described, as these may vary. It is also to be understood that the terminology used in the description is for the purposes of describing the particular versions or embodiments only, and is not intended to limit the scope of the present disclosure. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of embodiments of the present disclosure, the methods, devices, and materials in some embodiments are now described. All publications mentioned herein are incorporated by reference in their entirety. Nothing herein is to be construed as an admission that the present disclosure is not entitled to antedate such disclosure by virtue of prior invention.

Definitions

[0020] Unless otherwise defined herein, scientific and technical terms used in connection with the present disclosure shall have the meanings that are commonly understood by those of ordinary skill in the art. The meaning and scope of the terms should be clear. However, in the event of any latent ambiguity, definitions provided herein take precedent over any dictionary or extrinsic definition. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular.
[0021] The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified unless clearly indicated to the contrary. Thus, as a non-limiting example, a reference to “A and/or B,” when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A without B (optionally including elements other than B); in another embodiment, to B without A (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.
[0022] As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above unless context dictates otherwise. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.
[0023] The term “about” is used herein to mean within the typical ranges of tolerances in the art. For example, “about” can be understood as about 2 standard deviations from the mean. According to certain embodiments, when referring to a measurable value such as an amount and the like, “about” is meant to encompass variations of ±20%, ±10%, ±5%, ±1%, ±0.9%, ±0.8%, ±0.7%, ±0.6%, ±0.5%, ±0.4%, ±0.3%, ±0.2% or ±0.1% from the specified value as such variations are

appropriate to perform the disclosed methods. When “about” is present before a series of numbers or a range, it is understood that “about” can modify each of the numbers in the series or range.
[0024] The term “at least” prior to a number or series of numbers (e.g. “at least two”) is understood to include the number adjacent to the term “at least,” and all subsequent numbers or integers that could logically be included, as clear from context. When “at least” is present before a series of numbers or a range, it is understood that “at least” can modify each of the numbers in the series or range. Ranges provided herein are understood to include all individual integer values and all subranges within the ranges.
[0025] As used herein, the terms “comprising” (and any form of comprising, such as “comprise,” “comprises,” and “comprised”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”), or “containing” (and any form of containing, such as “contains” and “contain”), are inclusive or open-ended and do not exclude additional, unrecited elements or method steps.
[0026] As described herein, a “magnetic material” should be understood to include both one or more magnets and a material that is configured to magnetize in the presence of a magnet to thereby cause a magnetic attraction to said magnet. In various optional aspects, the magnet(s) can comprise one or more rare earth magnets, one or more ferrite magnets, one or more flexible magnets, or any other suitable magnet type (e.g., permanent magnets). The magnetic material that is configured to magnetize in the presence of a magnet can include, for example, a ferromagnetic material.
Systems
[0027] Referring to FIGS. **1-2**, the disclosure relates to a system **10** for detecting whether a projectile has passed through a conduit such as, for example, a hose. (As used herein, the terms “pellet” and “projectile” are interchangeably used. However, in alternative aspects, the projectile need not be a pellet.) For example, the system **10** can be used to perform a test including firing a pellet into the conduit and determining if the pellet passed through the conduit. This test can permit detecting defects such as a blockage within the interior of the conduit. In further aspects, a pellet or other projectile can be used to clean the conduit.
[0028] The system **10** comprises a housing **20** defining an inlet **22**. A trap **30** can be positioned within the housing **20**. The trap **30** can be moveable about and between a catching configuration (FIG. **6**) and a releasing configuration (FIG. **7**). When the trap **30** is in the catching configuration, a pellet cannot pass by the trap and is caught by the trap as it passes through the housing **20**. When the trap is in the releasing configuration, the pellet can pass by the trap.
[0029] The housing **20** can comprise a window **26** that permits visual detection of the trap **30** from a position outside of the housing. Accordingly, the window **26** can permit an operator to visually detect the presence or absence of a pellet trapped by the trap **30**. An actuator **40** can be configured to move the trap **30** about and between the catching configuration and the releasing configuration. For example, after detecting the presence of the pellet trapped by the trap **30**, the actuator **40** can be actuated to clear the trap **30** to prepare the system for a subsequent test.
[0030] In some aspects, the housing **20** can comprise a tubular structure. For example, in these aspects, the housing **20** can comprise cylindrical tubing **25**. In other aspects, the housing **20** can comprise rectangular tubing (i.e., having square or rectangular cross sections). In some optional aspects the window **26** can comprise a transparent tube. For example, the tubular structure can comprise at least one transparent tube segment. In some optional aspects, the housing **20** can comprise an elbow **27** positioned between the inlet **22** and the outlet **24**. For example, a first portion of the housing can extend along a horizontal axis (optionally, parallel to the horizontal axis), and a second portion can extend downwardly (optionally, parallel to a vertical axis), with the elbow between the first and second segments. In this way, gravity can permit the pellet to drop from the trap when the trap is moved to the releasing configuration. The housing **20** can extend along and define a projectile path of travel. In some optional aspects, the housing **20** can have a major cross sectional dimension (e.g., diameter) of at least 2 inches perpendicular to the pellet path

of travel. In some optional aspects, the housing **20** can have a major cross sectional dimension (e.g., diameter) of at least 3 inches.

[0031] Referring to FIG. **9**, in some aspects, the system **10** can comprise a retainer that secures or otherwise supports a conduit **110** so that an end **112***b* is in communication with the inlet **22** of the housing **22**. For example, the retainer can comprise a clamp **130**. In other aspects, the retainer can comprise a hook, a strap, a fastener, an adhesive, or any suitable structure for securing or otherwise supporting the conduit **110** relative to the housing **20**.

[0032] In some aspects, and as illustrated in FIGS. **3-4**, the trap **30** can be pivotable about a pivot axis **32** about and between the catching configuration (FIG. **6**) and the releasing configuration (FIG. **7**). The housing **20** can define an interior **28** through which the pellet travels. Optionally, the pivot axis **32** can extend through the interior **28** of the housing **20**. In some aspects, the trap **30** can be able to rotate 360 degrees about the pivotal axis. For example, it is contemplated that the trap **30** can be rotated (e.g., spun) about the axis **32** to release a pellet. In other aspects, rotation of the trap **30** can be inhibited past a rotational threshold. For example, in these aspects, rotation of the trap **30** can be limited to about a 90 degree pivot.

[0033] Referring to FIGS. **3-5**, in some aspects, the system **10** can comprise at least one magnet **34** that is configured to retain the trap in the catching configuration. In further aspects, the at least one magnet **34** can urge the trap **30** to return to the catching configuration. In this way, once the trap **30** is moved to release a pellet from the trap, the trap can then be reset to the catching configuration in preparation for catching the next pellet. In some aspects, the at least one magnet **34** can comprise a first magnet **34***a* that is coupled to the trap **30** and a second magnet **34***b* that is coupled to the housing **20**. In still further aspects, the system **10** can comprise one or a pair of first magnets **34***a* that are coupled to the trap **30** and a pair of second magnets **34***b* that are coupled to the housing **20**. The pair of first magnets **34***a* can be positioned on opposite sides of the trap (e.g., along an axis **12** that is perpendicular to the rotational axis **32** of the trap **30**). The pair of second magnets **34***b* can be positioned on opposite sides of the housing along the same axis **12** so that when the trap **30** is in the catching configuration, each magnet **34***a* of the first pair is positioned proximate to a respective magnet **24***b* of the second pair of magnets.

[0034] In other aspects, any combination of magnets and magnetizable material (e.g., ferromagnetic material that is configured to magnetize in the presence of magnets) that cooperate to attractively retain the trap **30** in the catching configuration can be used. Accordingly, the magnet(s) **34** and magnetic material can be positioned on the housing **20** and on the trap **30** so that the magnets cooperate to attractively retain the trap **30** in the catching configuration. For example, in some aspects, a magnet can be coupled to the housing, and a magnetizable material can be coupled to the trap **30** in respective positions so that the magnet and magnetic material are proximate to each other when the trap is in the catching configuration. In other aspects, a magnetizable material can be coupled to the housing, and a magnet can be coupled to the trap **30** in respective positions so that the magnet and magnetic material are proximate to each other when the trap is in the catching configuration.

[0035] In aspects in which the trap **30** can be rotated (e.g., spun) about the axis **32** to release a pellet, the at least one magnet **34** can stop the trap **30** from rotation in the catching configuration. In some aspects, the magnets **34** can be secured to the housing and/or the trap **30** via fasteners **35** (e.g., screws or bolts).

[0036] In some aspects, the trap **30** can be configured to permit gas to pass thereby. In this way, gas that drives the pellet through the conduit can be permitted to pass by the trap (with the trap in the catching configuration) while the pellet can be trapped. For example, in some embodiments, the trap **30** can comprise a screen. In other aspects, the trap **30** can comprise a plate **31**. In some optional aspects, the screen or the plate can be disc-shaped. For example, the disc-shaped screen or plate that generally corresponds to the cross-sectional shape of the interior **28** of the housing at the trap. In some aspects, the trap **30** can comprise one or a plurality of perforations. For example, in some aspects, the one or plurality of perforations can be formed through the trap **30**. In other aspects, the trap can comprise a screen or mesh that defines the plurality of perforations. In other aspects, the housing **20** and the trap **30** can cooperate to permit gas to pass by the trap with the trap in the catching configuration. For example, the trap **30** and the housing **20** can define a gap therebetween that can permit gas to pass by the trap with the trap in the catching configuration. Optionally, in these aspects, the gap can be defined between an interior surface of the housing **20** and the outer periphery of the trap **30**.

[0037] In additional or alternative aspects, and referring to FIG. **3**, the housing **20** can have an outer surface **58**. In some aspects, the housing **20** can define one or a plurality of perforations **54** along the circumferential surface. The one or a plurality of perforations **54** can permit gas to exhaust from the interior of the housing **20**. The one or a plurality of perforations **54** can be sized to inhibit the pellet from passing therethrough. In some optional aspects, the one or a plurality of perforations **54** can be positioned in or after the elbow **27** (where provided). In this way, the gas can be exhausted in a position along the path of the pellet after the pellet no longer needs the gas to carry the pellet through the housing **20**.

[0038] Referring to FIG. **1**, in some aspects, the system can comprise an object receptacle **50**. The object receptacle **40** can be, for example, a drum. In other aspects, the object receptacle **50** can be a bin, a box, a cage, a bucket, or any suitable receptacle. The object receptacle **50** can be configured to catch and hold pellets. The housing **20** can comprise an outlet **24** in communication with the object receptacle **50**. For example, the housing **20** can comprise a flange **29** that couples to the object receptacle **50**.

[0039] In some aspects, the object receptacle **50** can comprise at least one exhaust outlet that permits gas to pass therethrough. In this way, gas that drives the pellet through the conduit can be permitted to pass from the object receptacle **50** while retaining the pellet within the object receptacle. For example, in some aspects, the at least one exhaust outlet comprises a plurality of openings. Each opening can have a major dimension that is less than a predetermined dimension. For example, in some aspects, the openings can have a major dimension that is less about 1 cm, or less than about ½ cm. In this way, the openings can be sized to inhibit pellets from escaping therethrough.

[0040] In some aspects, a support element **52** can support at least a portion of the housing **20**. For example, the support element **52** can comprise a strut that couples to the object receptacle **50**.

[0041] In some aspects, the system **10** can comprise a conductor **56** (FIG. **5**) in communication with the trap **30** and/or the housing **20**. The conductor **56** can be configured to dissipate electrostatic energy to a ground. For example, it is contemplated that the pellet can accumulate static electricity as it travels through the conduit. The conductor **56** can dissipate at least a portion of the static electricity to inhibit the pellet from electrostatically adhering to the housing **20** or the trap **30**. The conductor **56** can be a wire or any suitable electrical conductor.

[0042] In some aspects, the housing **20** can comprise an access door **60** (shown in broken lines in FIG. **5**) proximate to the trap **30**. The access door **60** can permit a user to access the trap **30** from a position exterior to the housing **20**. In this way, an operator can clear any stuck pellet or refuse from the trap **30** as well as service the trap for any maintenance.

[0043] In some aspects, the actuator **40** can comprise a knob coupled to the trap **30**. For example, in some aspects, the actuator **40** can comprise a pair of knobs **42** coupled to the trap and positioned on opposite sides of the housing **20**. An operator can use one or more of the knobs **42** to rotate (optionally, spin) the trap **30** to release a pellet after confirming that the pellet was caught in the trap.

[0044] Referring to FIGS. **8**A and **8**B, in other aspects, the trap can comprise a hinged door **36**. In some aspects, the actuator **40** can comprise a lever **44**. In further aspects, the actuator can comprise a handle **46**. In exemplary aspects, the trap **30** can be spring-biased (e.g., via a spring **38**) toward the catching configuration. In some optional aspects, the trap **20** can comprise a pair of hinged doors.

[0045] In further aspects, the trap **30** can comprise a slidable gate. For example, the slidable gate can slide along a track. In other aspects, the trap **30** can comprise an iris.

[0046] Referring to FIG. **9**, the system **10** can further comprise a projectile launcher **100**. The projectile launcher **100** can use compressed gas from a compressed gas source to fire the pellet into a conduit **110** (e.g., tubing, such as, for example, hydraulic or pneumatic tubing). The projectile launcher **100** can be comprise a nozzle that is configured to be inserted into a first end **112***a* of the conduit **110**. An opposed second end **112***b* of the conduit **110** can be inserted into, or otherwise coupled to, the housing **20**. For example, the clamp **130** can secure the second end **112***b* to the housing **20** so that the second end of the conduit **110** is in communication with the inlet **22** of the housing. Optionally, the first end **112***a* of the conduit **110** can be coupled to the projectile launcher **100**. In other aspects, the conduit **110** need not be coupled to the projectile launcher **100**. Optionally, the second end **112***b* of the conduit **110** can be coupled to the housing **20**. In other aspects, the conduit **110** need not be coupled to the housing **20**.

[0047] In some aspects, the system **10** can be operated without need for any computing device. For example, an operator can visually inspect the trap **30** and operate the actuator **40** to release the pellet. In other aspects, and with reference to FIG. **10**, the system **10** can comprise an optical sensor **120** that is positioned to detect, through the window **26**, the presence or absence of an object trapped in the trap. For example, the optical sensor **120** can comprise a laser **122** and an optical detector **124** that is configured to detect emission from the laser. The laser and optical detector **124** can be positioned so that, when the pellet is trapped in the trap, the pellet blocks emission of light from the laser **122** to the optical detector **124**. The absence of detected light from the laser **122** can, therefore, correspond to detection of the pellet in the trap. In other aspects, other optical sensor **120** are contemplated, such as a motion sensor or an optical range detector.

[0048] The system **10** can further comprise a computing device **1001** that is in communication with the optical sensor **120**. In some aspects, the computing device **1001** can be configured to cause, upon receipt of a signal from the optical detector detecting the presence of the object trapped in the trap, the actuator **40** to move the trap to the releasing configuration. For example, the actuator can comprise a motor, a pneumatic actuator, or a solenoid in communication with the computing device **1001**.

[0049] In further aspects, the projectile launcher **1001** can be in communication with the computing device **1001**. In these aspects, the computing device **100** can be configured to initiate an alarm upon detecting an absence of the object trapped in the trap for a predetermined duration from operation of the pneumatic pellet launcher. For example, in some aspects, projectile launcher **1001** can send a signal to the computing device **1001** that a pellet has been fired, and the computing device can initiate a timer. Upon expiration of the timer, if the optical sensor **120** does not detect a pellet, the computing device **1001** can initiate the alarm (e.g., providing an error notification to an operator, turning on an audible or visual alarm, or preventing advancement of a test until an operator clears the alarm). In other aspects, the computing device can cause the pneumatic pellet launcher **100** to fire the projectile and initiate the timer based on a time at which the computing device caused the pneumatic pellet launcher **100** to fire the projectile.

Method

[0050] A method of using the system **10** can comprise coupling a first end of a conduit to a projectile launcher and positioning a second end of the conduit at the inlet of the housing. A pellet can be propelled through the conduit.

[0051] The method can further comprise activating movement of the actuator to move the trap to the releasing configuration upon visually detecting the pellet being trapped in the trap. In aspects in which the actuator comprises a knob coupled to the trap, the knob can be spun to rotate the trap. In some aspects, a determination that the pellet is stuck within the hose can be made by visual inspection of the trap.

[0052] In some aspects, visually detecting the presence or absence of a pellet being trapped in the trap can be performed by an operator. In other aspects, visually detecting the presence or absence of a pellet being trapped in the trap can be performed by an optical sensor **120**.
[0053] In aspects in which the system comprises an object receptacle **40**. One or more pellets can be returned from the object receptacle **40** to the projectile launcher **100** for reuse.
[0054] In aspects in which the housing **20** comprises an access door **60** proximate to the trap **30**, the access door can be opened, and a stuck pellet can be cleared from the housing **20**.
Computing Device
[0055] FIG. **11** shows an exemplary computing system **1000** including an exemplary configuration of a computing device **1001** that can be used with the system **10** (FIG. **10**). The computing device **1001** can be configured to receive data such as, but not limited to, sensor data from the optical sensor **120** (FIG. **10**).
[0056] The computing device **1001** may comprise one or more processors **1003**, a system memory **1012**, and a bus **1013** that couples various components of the computing device **1001** including the one or more processors **1003** to the system memory **1012**. In the case of multiple processors **1003**, the computing device **1001** may utilize parallel computing.
[0057] The bus **1013** may comprise one or more of several possible types of bus structures, such as a memory bus, memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures.
[0058] The computing device **1001** may operate on and/or comprise a variety of computer readable media (e.g., non-transitory). Computer readable media may be any available media that is accessible by the computing device **1001** and comprises, non-transitory, volatile and/or non-volatile media, removable and non-removable media. The system memory **1012** has computer readable media in the form of volatile memory, such as random access memory (RAM), and/or non-volatile memory, such as, but not limited to read only memory (ROM).
[0059] The computing device **1001** may also comprise other removable/non-removable, volatile/non-volatile computer storage media. The mass storage device **1004** may provide non-volatile storage of computer code, computer readable instructions, data structures, program modules, and other data for the computing device **1001**. The mass storage device **1004** may be a hard disk, a removable magnetic disk, a removable optical disk, magnetic cassettes or other magnetic storage devices, flash memory cards, CD-ROM, digital versatile disks (DVD) or other optical storage, random access memories (RAM), read only memories (ROM), electrically erasable programmable read-only memory (EEPROM), and the like.
[0060] Any number of program modules may be stored on the mass storage device **1004**. An operating system **1005** and projectile detection software **1006** may be stored on the mass storage device **1004**. One or more of the operating system **1005** and/or projectile detection software **1006** (or some combination thereof) may comprise program modules and the projectile detection software **1006**. Optical sensor data **1007** may also be stored on the mass storage device **1004**. The optical sensor data **1007** may be stored in any of one or more databases known in the art. The databases may be centralized or distributed across multiple locations within the network **1015**.
[0061] In some aspects, the computing device **1001** (server) may be a cloud-based or web-based server without departing from a broader scope of the present disclosure. In some aspects, the remote computing device **1014** may include an implementation of a client instance of the computing device **1001**. As such, a user may interact with the computing device **1001** through the remote computing device **1014**, e.g., the client instance implemented therein. In some aspects, the remote computing device **1014** may include processors, memory, display interfaces/devices, other output devices, sensors, features of the measuring device, etc., without departing from a broader scope of the present disclosure.
[0062] In some aspects, a user may enter commands and information into the computing device **1001** using an input device. Such input devices comprise, but are not limited to, a joystick, a

touchscreen display, a keyboard, a pointing device (e.g., a computer mouse, remote control), a microphone, a scanner, tactile input devices such as gloves, and other body coverings, motion sensor, speech recognition, and the like. These and other input devices may be connected to the one or more processors **1003** using a human machine interface **1002** that is coupled to the bus **1013**, but may be connected by other interface and bus structures, such as a parallel port, game port, an IEEE 1394 Port (also known as a Firewire port), a serial port, network adapter **1008**, and/or a universal serial bus (USB).

[0063] A display device **1011** may also be connected to the bus **1013** using an interface, such as a display adapter **1009**. It is contemplated that the computing device **1001** may have more than one display adapter **1009** and the computing device **1001** may have more than one display device **1011**. A display device **1011** may be a monitor, an LCD (Liquid Crystal Display), light emitting diode (LED) display, television, smart lens, smart glass, and/or a projector. In addition to the display device **1011**, other output peripheral devices may comprise components such as speakers (not shown) and a printer (not shown) which may be connected to the computing device **1001** using Input/Output Interface **1010**. Any step and/or result of the methods may be output (or caused to be output) in any form to an output device. In some aspects, any appropriate output from the computing device **1001** may be transmitted to the second computing device **30** and/or the remote computing device **1014** for presentation to a user via the client instance of the computing device **1001**. Such output may be any form of visual representation, including, but not limited to, textual, graphical, animation, audio, tactile, and the like. The display device **1011** and computing device **1001** may be part of one device, or separate devices.

[0064] The computing device **1001** may operate in a networked environment using logical connections to one or more remote computing devices **1014***a,b,c*. The other remote computing devices **1014***a,b,c* may be a personal computer, computing station (e.g., workstation), portable computer (e.g., laptop, mobile phone, tablet device), smart device (e.g., smartphone, smart watch, activity tracker, smart apparel, smart accessory), security and/or monitoring device, a server, a router, a network computer, a peer device, or other common network node, and so on. Logical connections between the computing device **1001** and the remote computing devices may be made using a network **1015**, such as a local area network (LAN) and/or a general wide area network (WAN), or a Cloud-based network. Such network connections may be through a network adapter **1008**. A network adapter **1008** may be implemented in both wired and wireless environments. Such networking environments are conventional and commonplace in dwellings, offices, enterprise-wide computer networks, intranets, and the Internet. It is contemplated that the remote computing devices can optionally have some or all of the components disclosed as being part of computing device **1001**. In various further aspects, it is contemplated that some or all aspects of data processing described herein can be performed via cloud computing on one or more servers or other remote computing devices **1014**. Accordingly, at least a portion of the system **1000** can be configured with internet connectivity.

[0065] It should be noted that, the above embodiments are only intended for describing the present disclosure, and should not be interpreted as limitation to the technical solutions of the present disclosure. Although the present disclosure is described in detail in conjunction with the above embodiments, it should be understood by the skilled in the art that, modifications or equivalent substitutions may still be made to the present disclosure by those skilled in the art; and any technical solutions and improvements thereof without departing from the spirit and scope of the present disclosure also fall into the scope of the present disclosure defined by the claims.

EXEMPLARY ASPECTS

[0066] In view of the described products, systems, and methods and variations thereof, herein below are described certain more particularly described aspects of the invention. These particularly recited aspects should not however be interpreted to have any limiting effect on any different claims containing different or more general teachings described herein, or that the "particular" aspects are somehow limited in some way other than the inherent meanings of the language literally used therein.

[0067] Aspect 1: A system comprising: [0068] a housing defining an inlet; [0069] a trap positioned within the housing, the trap moveable about and between a catching configuration and a releasing configuration; and [0070] an actuator that is configured to move the trap about and between the catching configuration and the releasing configuration, [0071] wherein the housing comprises a window that permits visual detection of the trap from a position outside of the housing.

[0072] Aspect 2: The system of aspect 1, wherein the trap is pivotable about a pivot axis about and between the catching configuration and the releasing configuration.

[0073] Aspect 3: The system of aspect 2, wherein the housing defines an interior, wherein the pivot axis extends through the interior of the housing.

[0074] Aspect 4: The system of aspect 2 or aspect 3, wherein the trap is rotatable about 360 degrees about the pivotal axis.

[0075] Aspect 5: The system of any one of the preceding aspects, further comprising at least one magnet that is configured to retain the trap in the catching configuration.

[0076] Aspect 6: The system of aspect 5, wherein the at least one magnet comprises at least one first magnet that is coupled to the trap and at least one second magnet that is coupled to the housing.

[0077] Aspect 7: The system of aspect 6, wherein the at least one first magnet comprises a first pair of magnets that are positioned on opposite sides of the trap, wherein the at least one second magnet comprises a second pair of magnets that are positioned on opposite sides of the housing so that when the trap is in the catching configuration, each magnet of the first pair is positioned proximate to a respective magnet of the second pair of magnets.

[0078] Aspect 8: The system of any one of the preceding aspects, wherein the trap comprises a screen or a plate.

[0079] Aspect 9: The system of aspect 8, wherein the plate is disc shaped.

[0080] Aspect 10: The system of any one of the preceding aspects, wherein the trap comprises one or a plurality of perforations to permit air flow therethrough when the trap is in the catching configuration.

[0081] Aspect 11: The system of any one of the preceding aspects, further comprising an object receptacle, wherein the housing comprises an outlet in communication with the object receptacle.

[0082] Aspect 12: The system of aspect 11, wherein the object receptacle comprises at least one exhaust outlet that permits gas to pass therethrough.

[0083] Aspect 13: The system of aspect 12, wherein the at least one exhaust outlet comprises a plurality of openings, each opening having a major dimension that is less than about 1 cm.

[0084] Aspect 14: The system of any one of aspects 11-13, wherein the object receptacle comprises a drum.

[0085] Aspect 15: The system of any one of the preceding aspects, wherein the housing comprises a tubular structure.

[0086] Aspect 16: The system of any one of aspects 12-15, wherein the window comprises a transparent tube.

[0087] Aspect 17: The system of any one of the preceding aspects, wherein the housing comprises: [0088] an outlet; and [0089] an elbow positioned between the inlet and the outlet.

[0090] Aspect 18: The system of aspect 16, wherein the housing comprises cylindrical tubing.

[0091] Aspect 18: The system of aspect 16, wherein the housing comprises rectangular tubing.

[0092] Aspect 20: The system of any one of the preceding aspects, wherein the housing has a major cross sectional dimension of at least about 3 inches.

[0093] Aspect 21: The system of any one of the preceding aspects, further comprising a conductor in electrical communication with the trap and configured to dissipate electrostatic energy to a ground.

[0094] Aspect 222: The system of any one of the preceding aspects, wherein the housing comprises an access door proximate to the trap, wherein the access door permits a user to access the trap from a position exterior to the housing.
[0095] Aspect 23: The system of any one of the preceding aspects, wherein the housing has an interior and an outer surface, wherein the housing defines one or a plurality of perforations extending from the interior to the outer surface.
[0096] Aspect 24: The system of any one of the preceding aspects, wherein the actuator comprises a knob coupled to the trap.
[0097] Aspect 25: The system of aspect 24, wherein the actuator comprises a pair of knobs coupled to the trap and positioned on opposite sides of the housing.
[0098] Aspect 26: The system of any one of the preceding aspects, wherein the trap comprises a hinged door.
[0099] Aspect 27: The system of aspect 26, wherein the trap comprises a pair of hinged doors.
[0100] Aspect 28: The system of any one of the preceding aspects, wherein the trap comprises a slidable gate.
[0101] Aspect 29: The system of any one of the preceding aspects, wherein the trap comprises an iris.
[0102] Aspect 30: The system of any one of the preceding aspects, wherein the actuator comprises a lever.
[0103] Aspect 31: The system of any one of the preceding aspects, wherein the trap is spring-biased toward the catching configuration.
[0104] Aspect 32: The system of any one of the preceding aspects, wherein the actuator comprises a handle.
[0105] Aspect 33: The system of aspect 1, further comprising a projectile launcher.
[0106] Aspect 34: The system of any one of the preceding aspects, further comprising an optical sensor positioned to detect, through the window, the presence or absence of an object trapped in the trap.
[0107] Aspect 35: The system of aspect 34, further comprising a computing device in communication with the optical sensor and the actuator, wherein the computing device is configured to cause, upon receipt of a signal from the optical detector detecting the presence of the object trapped in the trap, the actuator to move the trap to the releasing configuration.
[0108] Aspect 36: The system of aspect 35, wherein the actuator comprises a motor or a solenoid.
[0109] Aspect 37: The system of aspect 35, further comprising a projectile launcher in communication with the computing device, wherein the computing device is configured to initiate an alarm upon detecting an absence of the object trapped in the trap for a predetermined duration from operation of the projectile launcher.
[0110] Aspect 38: A method of using the system of any one of the preceding aspects, the method comprising: [0111] coupling a first end of a conduit to a projectile launcher; [0112] positioning a second end of the conduit at the inlet of the housing; and [0113] propelling, using the projectile launcher, a pellet through the conduit.
[0114] Aspect 39: The method of aspect 38, further comprising activating movement of the actuator to move the trap to the releasing configuration upon visually detecting the pellet being trapped in the trap.
[0115] Aspect 40: The method of aspect 38 or aspect 39, wherein the actuator comprises a knob coupled to the trap; and wherein activating movement of the actuator comprises spinning the knob to rotate the trap.
[0116] Aspect 41: The method of any one of aspects 38-40, wherein the trap is configured to automatically return to the catching configuration if the trap is positioned in the releasing configuration.
[0117] Aspect 42: The method of any one of aspects 38-41, further comprising a step of

determining whether the pellet is stuck within the hose by visual inspection of the trap.
[0118] Aspect 43: The method of any one of aspects 38-42, wherein the system further comprises an object receptacle, wherein the housing comprises an outlet in communication with the object receptacle, the method comprising returning at least one pellet from the object receptacle to the projectile launcher for reuse.
[0119] Aspect 44: The method of any one of aspects 38-43, wherein the housing comprises an access door proximate to the trap, wherein the access door permits a user to access the trap from a position exterior to the housing, the method further comprising: [0120] opening the access door; and [0121] clearing a stuck pellet from the housing.
[0122] All referenced journal articles, patents, and other publications are incorporated by reference herein in their entireties.

Claims

1. A system comprising: a housing defining an inlet; a trap positioned within the housing, the trap moveable about and between a catching configuration and a releasing configuration; and an actuator that is configured to move the trap about and between the catching configuration and the releasing configuration, wherein the housing comprises a window that permits visual detection of the trap from a position outside of the housing.
2. The system of claim 1, wherein the trap is pivotable about a pivot axis about and between the catching configuration and the releasing configuration.
3. The system of claim 2, wherein the housing defines an interior, wherein the pivot axis extends through the interior of the housing.
4. The system of claim 2, wherein the trap is rotatable about 360 degrees about the pivotal axis.
5. The system of claim 1, further comprising at least one magnet that is configured to retain the trap in the catching configuration.
6. The system of claim 5, wherein the at least one magnet comprises at least one first magnet that is coupled to the trap and at least one second magnet that is coupled to the housing.
7. The system of claim 6, wherein the at least one first magnet comprises a first pair of magnets that are positioned on opposite sides of the trap, wherein the at least one second magnet comprises a second pair of magnets that are positioned on opposite sides of the housing so that when the trap is in the catching configuration, each magnet of the first pair is positioned proximate to a respective magnet of the second pair of magnets.
8. The system of claim 1, wherein the trap comprises a screen or a plate.
9. The system of claim 8, wherein the plate is disc shaped.
10. The system of claim 1 further comprising an object receptacle, wherein the housing comprises an outlet in communication with the object receptacle.
11. The system of claim 10, wherein the object receptacle comprises at least one exhaust outlet that permits gas to pass therethrough.
12. The system of claim 1, wherein the housing comprises a tubular structure.
13. The system of claim 12, wherein the window comprises a transparent tube.
14. The system of claim 1, wherein the housing comprises: an outlet; and an elbow positioned between the inlet and the outlet.
15. The system of claim 1, wherein the housing has an interior and an outer surface, wherein the housing defines one or a plurality of perforations extending from the interior to the outer surface.
16. The system of claim 1, wherein the actuator comprises a knob coupled to the trap
17. The system of claim 1 further comprising a projectile launcher.
18. The system of claim 1 further comprising: an optical sensor positioned to detect, through the window, the presence or absence of an object trapped in the trap; and a computing device in communication with the optical sensor and the actuator, wherein the computing device is

configured to cause, upon receipt of a signal from the optical detector detecting the presence of the object trapped in the trap, the actuator to move the trap to the releasing configuration.

19. The system of claim 18, wherein the actuator comprises a motor or a solenoid.

20. The system of claim 18, further comprising a projectile launcher in communication with the computing device, wherein the computing device is configured to initiate an alarm upon detecting an absence of the object trapped in the trap for a predetermined duration from operation of the projectile launcher.